

Product Information

ISSUE Date : 2004-12-04

MODEL : LTV350QV-F02

Note : The Product and specifications are subject to change without any notice.

Please ask for the latest Product Standards to guarantee the satisfaction of your product requirements.

PREPARED BY : LCD Mobile Display Development

Samsung Electronics Co. , LTD.

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Revision History

Date	Rev. No.	Page	Summary
Nov. 22. 2004	000		Rev.000 was issued.
Dec. 4. 2004	001	5	Vertical Module size was revised.
		23,24	Module weight was added.
			Power on sequence was revised.

General Description

* Description

LTV350QV-F02 is a TMR(Transmissive with Micro Reflective) type color active matrix TFT (Thin Film Transistor) liquid crystal display (LCD) that uses amorphous silicon TFT as a switching devices. This model is composed of a TFT-LCD module, a driver circuit and a back-light unit.

The resolution of a 3.5" contains 320RGBx240 dots and can display up to 16.7M colors.

* Features

- Transmissive with Micro Reflective type and back-light with six LEDs are available.
- Visible in outdoor & back-light off condition with reflectivity.
- TN(Twisted Nematic) mode.
- Line inversion mode with stripe type.
- 24bit RGB Interface + Serial Peripheral Interface(SPI)
- DE(Data Enable, Dotclk) mode, SYNC(Vsync, Hsync, Dotclk) mode

* Applications

- Display terminals for DSC(Digital Still Camera), PMP(Portable Multimedia Player) application products.

* General information

Items	Specification	Unit	Note
Display area	70.08(H) x 52.56(V)	mm	-
Driver element	a-Si TFT active matrix	-	-
Display colors	16.7M	colors	-
Number of pixels	320RGB(H) x 240(V)	dot	-
Pixel arrangement	stripe type	-	-
Pixel pitch	0.219(H) x 0.219(V)	mm	-
Display mode	Normally White	-	-
Viewing direction	6	o'clock	-

* Mechanical information

Item		Min.	Typ.	Max.	Unit	Note
Module size	Horizontal(H)	76.70	76.90	77.10	mm	-
	Vertical(V)	63.70	63.90	64.10	mm	(1)
	Depth(D)	2.85	3.20	3.25	mm	(1)
Weight		-	31.75		g	-

Note (1) Not include FPC

Refer to the Outline Dimension in the "12.Outline Dimension" for further information.

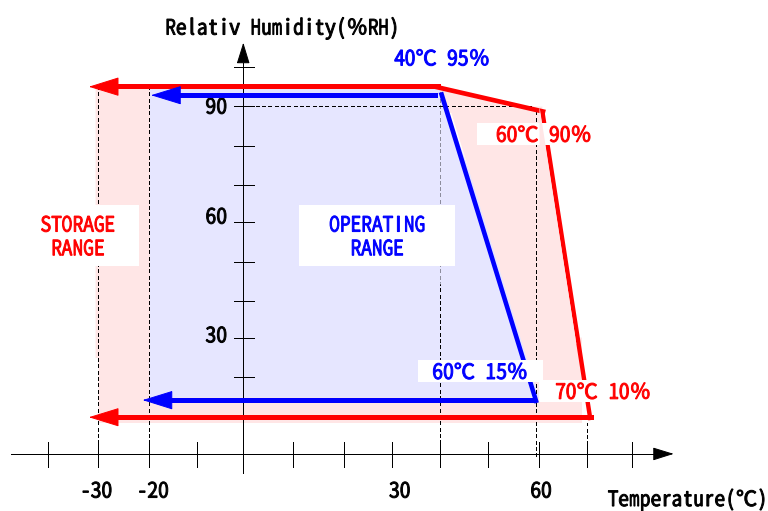
1. Absolute Maximum Ratings

1.1 Absolute Ratings of Environment

Item	Symbol	Min.	Max.	Unit	Note
Storage temperature	T _{STG}	-30	70	°C	(1)
Operating temperature (Ambient temperature)	T _{OPR}	-20	60	°C	(1),(2)

Note (1) 95 % RH Max. (40 °C ≥ Ta)

Maximum wet-bulb temperature at 39 °C or less. (Ta > 40 °C) No condensation.



(2) In case of below 0°, the response time of liquid crystal (LC) becomes slower and the color of panel becomes darker than normal one.

Level of retardation depends on temperature, because of LC's characteristics.

1.2 Electrical Absolute Ratings

(1) TFT-LCD Module

(Ta = 25 ± 2°C, V_{ss}=GND=0)

Characteristics	Symbol	Min.	Max.	Unit	Note
Digital supply voltage	V _{CC}	-0.3	4.3	V	-
Analog supply voltage	DDVDH	-0.3	6.0	V	-
Gate On voltage	V _{GH}	-0.3	22.0	V	-
Gate Off voltage	V _{GL}	-18.5	0.3	V	-

(2) Back-Light Unit

(Ta = 25 ± 2°C)

Item	Symbol	Min.	Max.	Unit.	Note
Current	I _B		25	mA	(1)

Note (1) Permanent damage to the device may occur if maximum values are exceeded or reverse voltage is loaded.

Functional operation should be restricted to the conditions described under normal operating conditions.

2. Optical Characteristics

The following items are measured under stable conditions. The optical characteristics should be measured in a dark room or equivalent state with the methods shown in Note (1).

Measuring equipment: LCD-5000, BM-5A, BM-7, PR-650, EZ-Contrast

(Ta = 25 ± 2°C, V_{CC} = V_{CI} = 3.3V, I_B = 20mA)

Item		Symbol	Condition	Min.	Typ.	Max.	Unit	Note	
Contrast ratio (Center point)		C/R	NOTE (1) $\phi = 0$ $\theta = 0$ Normal Viewing Angle B/L On	150	300	-	-	(2) BM-5A	
Luminance of white (Center point)		YL		150	200	-	cd/m ²	(3) BM-5A	
Response time	Rising:Tr	Tr+Tf		$\phi = 0$ $\theta = 0$	-	25	30	msec	(4) BM-7
	Falling:Tf								
Color chromaticity (CIE 1931)	White	Wx		Normal				-	(5) PR-650
		Wy							
	Red	Rx		Viewing Angle					
		Ry							
	Green	Gx		B/L On					
		Gy							
	Blue	Bx							
		By							
Viewing angle	Hor.	θ L	C/R $\geq$ 10 B/L On				Degrees	(6) Ez-Contrast	
		θ R							
	Ver.	ϕ H							
		ϕ L							

T.B.D.

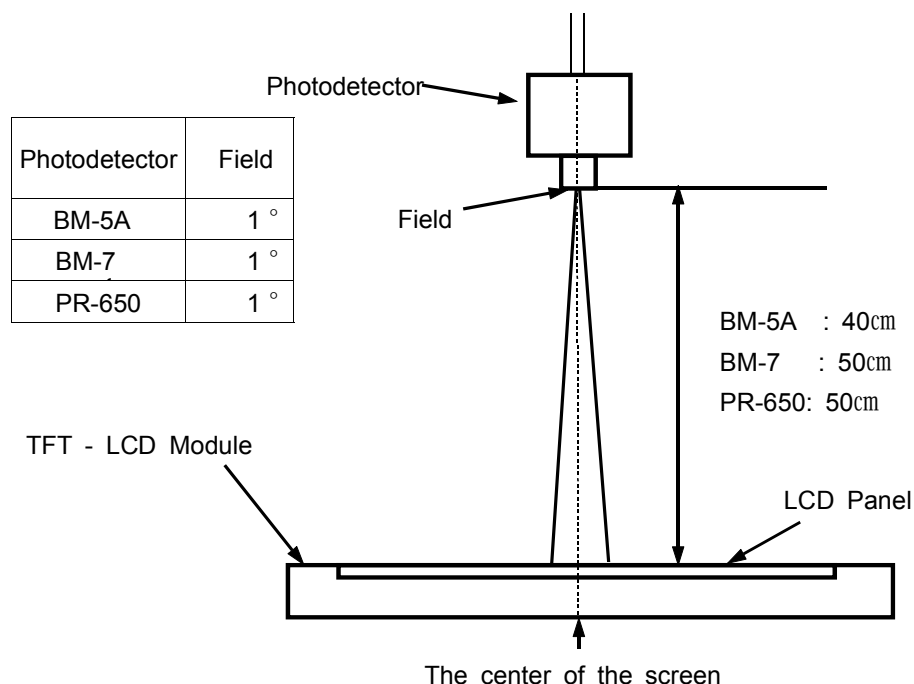
T.B.D.

Note (1) Test Equipment Setup

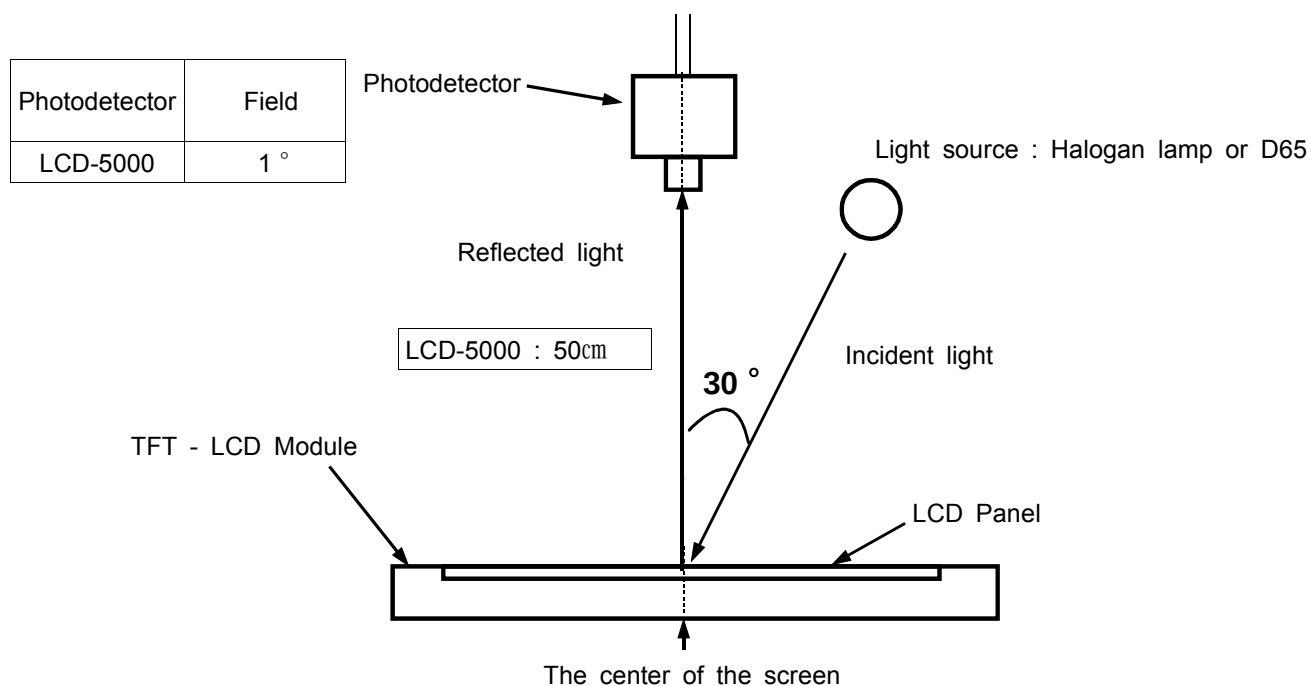
After stabilizing and leaving the panel alone at a given temperature for 30 min , the measurement should be executed. Measurement should be executed in a stable, windless, and dark room. 30 min after lighting the back-light. This should be measured in the center of screen.

Environment condition : $T_a = 25 \pm 2 \text{ }^{\circ}\text{C}$

Back-Light On condition



Back-Light Off condition



Optical Measuring Equipment Setup

Note (2) Definition of Contrast Ratio (C/R) : Ratio of gray max (Gmax) & gray min (Gmin) at the center point

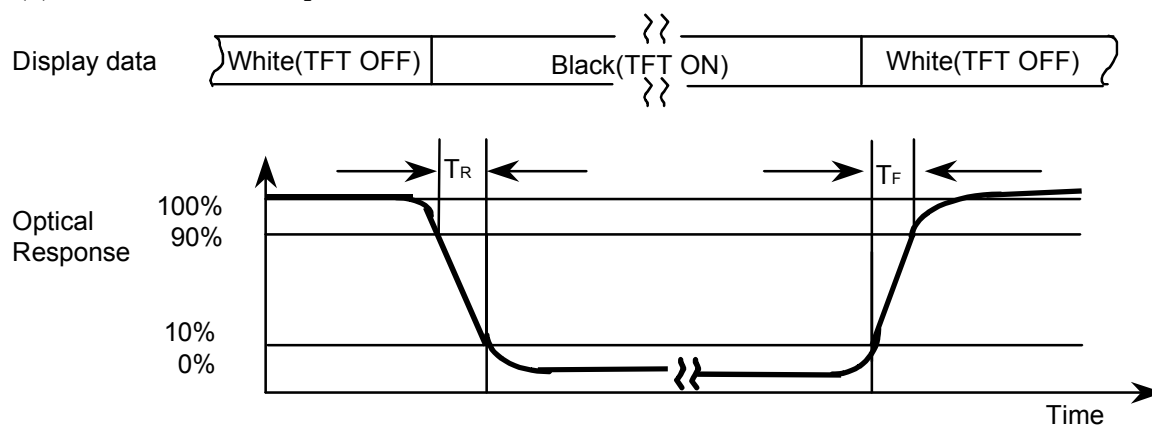
$$CR = \frac{G_{\max}}{G_{\min}}$$

* Gmax : Luminance with all pixels white

* Gmin : Luminance with all pixels black

Note (3) Definition of Luminance of White : Luminance of white at the center point

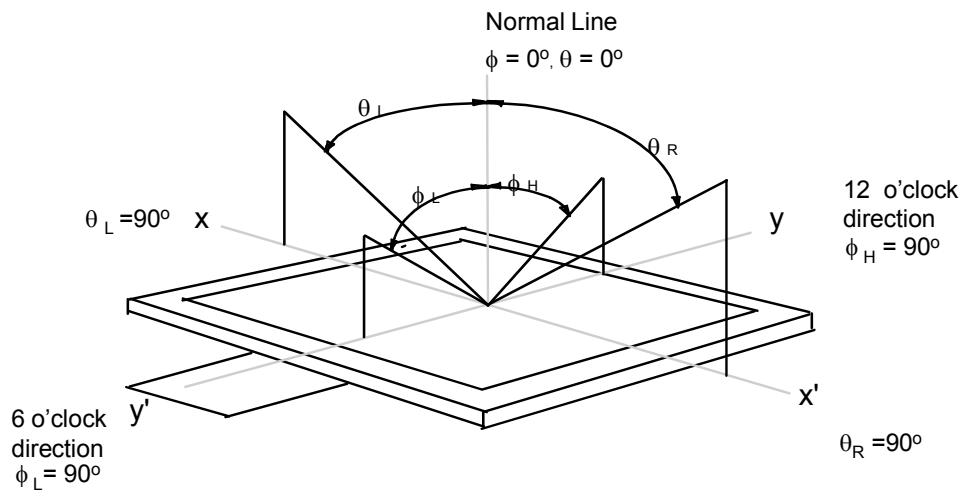
Note (4) Definition of Response time : Sum of T_r , T_f



Note (5) Definition of Color Chromaticity (CIE 1931)

Color coordinate of white & red, green, blue at center point.

Note (7) Definition of Viewing Angle : Viewing angle range ($CR \geq 10$)



3. Electrical Characteristics

3.1 TFT-LCD Module

Ta = 25 ± 2°C

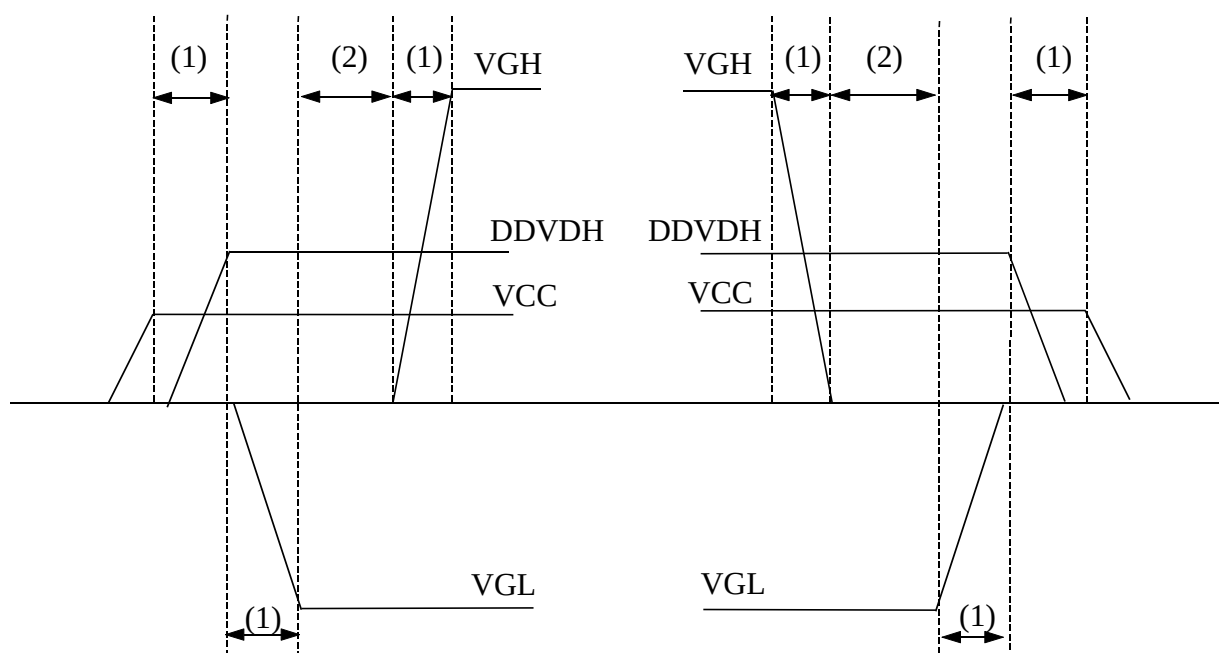
Characteristics		Symbol	Min.	Typ.	Max.	Unit	Note
Digital supply voltage		V _{CC}	2.5	3.3	3.3	V	-
Analog supply voltage		DDVDH	4.8	5.0	5.0	V	-
Gate On voltage		V _{GH}	17.0	18.0	19.0	V	-
Gate Off voltage		V _{GL}	-11	-10	-9.0	V	-
Power Dissipation	Full	P _{FULL}	TBD	TBD	TBD	mW	(1),(2)
Frame frequency		f _{Frame}	60	60	90	Hz	-
Dot Clock		DOTCLK	-	-	30.0	MHz	-
Serial Clock		SCL	-	-	16.0	MHz	-

* To prevent a latch-up or DC operation of the LCD module, the power on/off sequence should be as the Chapter 9. Power On/Off Sequence.

Note (1) V_{CC} = 3.3V, f_{Frame} = 60Hz, DOTCLK = 14.4MHz

(2) Dissipation current check pattern

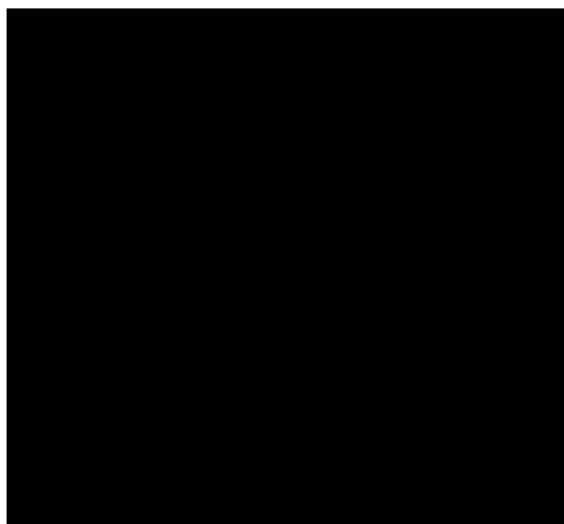
- Power On : V_{CC} → DDVDH → V_{GL} → V_{GH} → Data
- Power Off : Data → V_{GH} → V_{GL} → DDVDH → V_{CC}



(1) = T.B.D.

(2) = T.B.D.

▶ 0 Gray black pattern



3.2 Back-Light unit

The back-light system is an edge-lighting type with **six** white LED(Light Emitting Diode)s.

(Ta=25 ± 2°C)

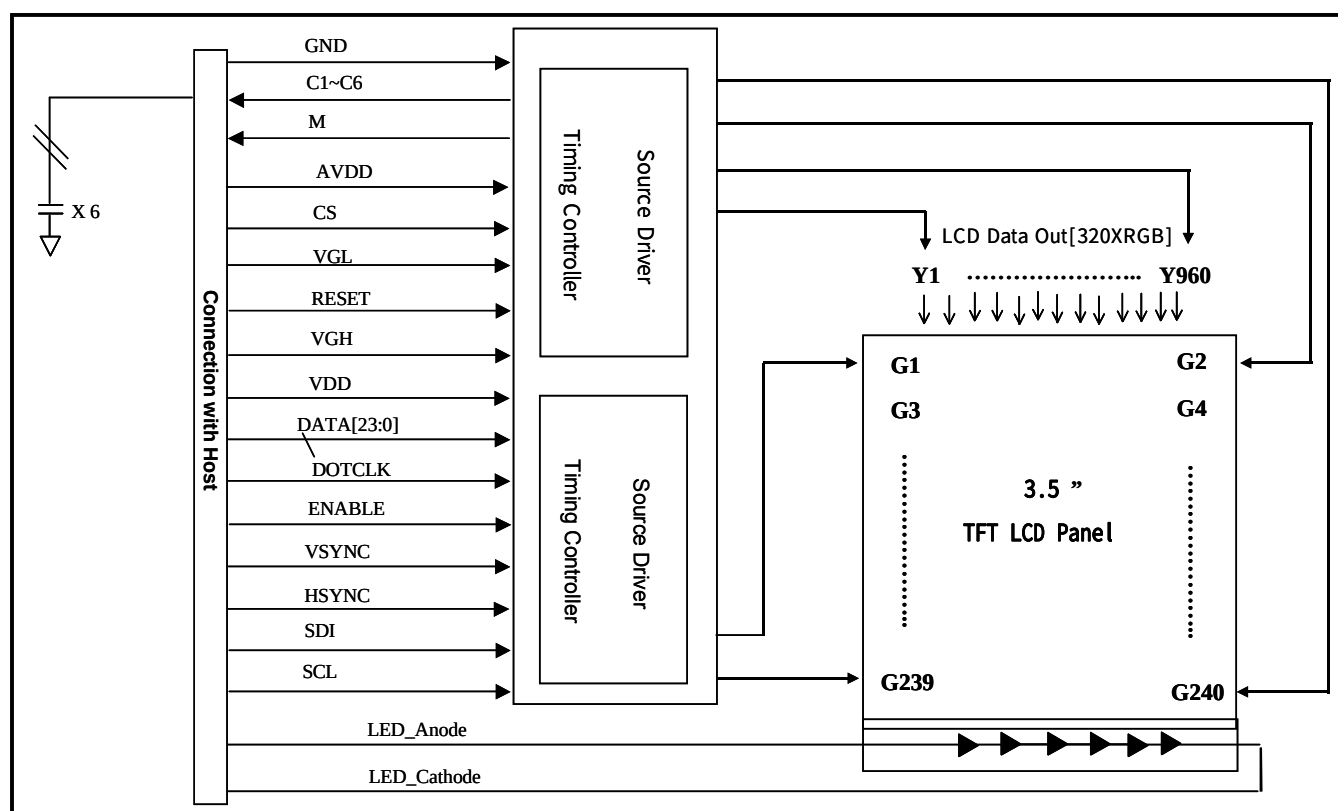
Item	Symbol	Min.	Typ.	Max.	Unit	Note
Current	I _B	-	20	-	mA	(1)
Power Consumption	P _{BL}	-	TBD	-	mW	(2)

Note (1) **Six** LEDs serial type.

(2) Where I_B = 20mA, V_B = P_{BL} / I_B

4. Block Diagram

4.1 TFT-LCD Module (Interface System Structure) with Back Light Unit



5. Input Terminal Pin Assignment

5.1 Input Signal & Power (Connector type : 54Pin / 0.5mm pitch / Bottom contact)
 - kyocera elco : 6240 series

No	Symbol	Description	I/O	No	Symbol	Description	I/O
1,2	LED_Cathode	LED_Cathode	I	27	DATA15	Green Data(MSB)	I
3,4	LED_Anode	LED_Anode	I	28	DATA16	Red Data (LSB)	I
5	C6	Stable Capacitor connection	O	29	DATA17	Red Data	I
6	C5	Stable Capacitor connection	O	30	DATA18	Red Data	I
7	M	The signal to generate Vcom	O	31	DATA19	Red Data	I
8	RESET	Reset	I	32	DATA20	Red Data	I
9	CS	Chip Select	I	33	DATA21	Red Data	I
10	SCL	Serial Clock	I	34	DATA22	Red Data	I
11	SDI	Serial Data	I	35	DATA23	Red Data(MSB)	I
12	DATA0	Blue Data (LSB)	I	36	HSYNC	Horizontal Synchronous Signal	I
13	DATA1	Blue Data	I	37	VSYNC	Vertical Synchronous Signal	I
14	DATA2	Blue Data	I	38	DOTCLK	Data Colck	I
15	DATA3	Blue Data	I	39,40	AVDD	Vanalog	I
16	DATA4	Blue Data	I	41,42	VDD	Vdigital	I
17	DATA5	Blue Data	I	43	C4	Stable Capacitor connection	O
18	DATA6	Blue Data	I	44,45	VGL	Vgoff Voltage	I
19	DATA7	Blue Data(MSB)	I	46	C3	Stable Capacitor connection	O
20	DATA8	Green Data(LSB)	I	47	VGH	Vgon Voltage	I
21	DATA9	Green Data	I	48	C2	Stable Capacitor connection	O
22	DATA10	Green Data	I	49	C1	Stable Capacitor connection	O
23	DATA11	Green Data	I	50,51	VCOM	VCOM	I
24	DATA12	Green Data	I	52	ENABLE	Data enabling signal	I
25	DATA13	Green Data	I	53,54	GND	Ground	I
26	DATA14	Green Data	I				

Note (1) Connect Hsync, Vsync pin at Vcc or GND in DE mode.

(2) Connect ENABLE pin at Vcc or GND in SYNC mode.

5.2 Input Signal, Basic Display Colors and Gray Scale of Each Colors

COLOR	DISPLAY	DATA SIGNAL																											GRAY SCALE LEVEL
		RED									GREEN								BLUE										
		R0	R1	R2	R3	R4	R5	R6	R7	G0	G1	G2	G3	G4	G5	G6	G7	B0	B1	B2	B3	B4	B5	B6	B7				
BASIC COLOR	BLACK	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-		
	BLUE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	-		
	GREEN	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-		
	CYAN	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	-		
	RED	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-		
	MAGENTA	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	-		
	YELLOW	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-		
	WHITE	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	-		
GRAY SCALE OF RED	BLACK	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	R0		
	DARK ↑	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	R1		
		0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	R2		
		:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	R3~R252		
		:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:			
	LIGHT ↓	1	0	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	R253		
		0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	R254		
	RED	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	R255		
GRAY SCALE OF GREEN	BLACK	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	G0		
	DARK ↑	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	G1		
		0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	G2		
		:			:	:	:	:	:	:			:	:	:	:	:	:			:	:	:	:	:	:	G3~G252		
		:			:	:	:	:	:	:			:	:	:	:	:	:			:	:	:	:	:	:			
	LIGHT ↓	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	G253		
		0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	G254		
	GREEN	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	G255		
GRAY SCALE OF BLUE	BLACK	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	B0		
	DARK ↑	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	B1		
		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	B2		
		:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	B3~B60		
		:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:			
	LIGHT ↓	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1	1	1	B61		
		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	B62		
	BLUE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	B63		

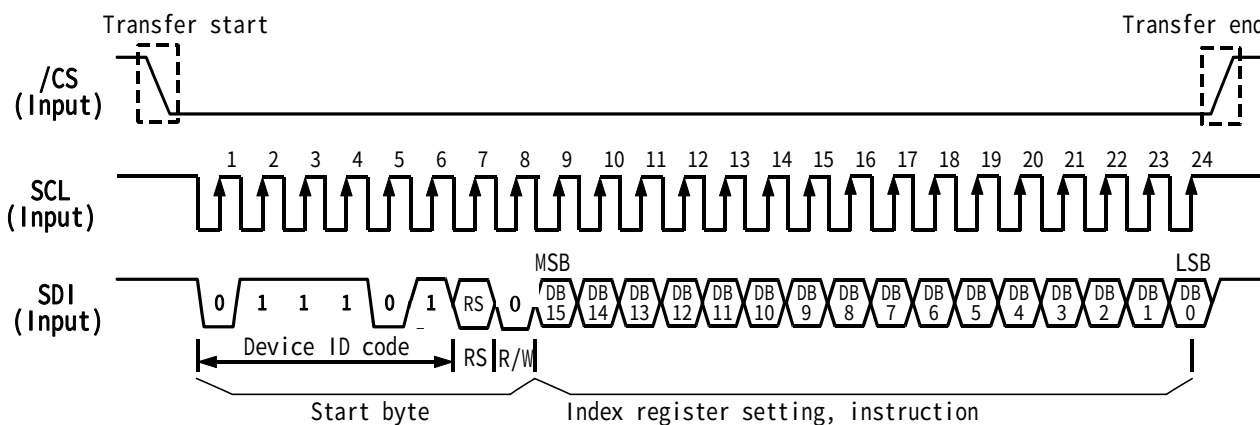
Note) Definition of Gray :

Rn : Red Gray, Gn : Green Gray, Bn : Blue Gray (n = Gray level)

Input Signal : 0 = Low level voltage, 1 = High level voltage

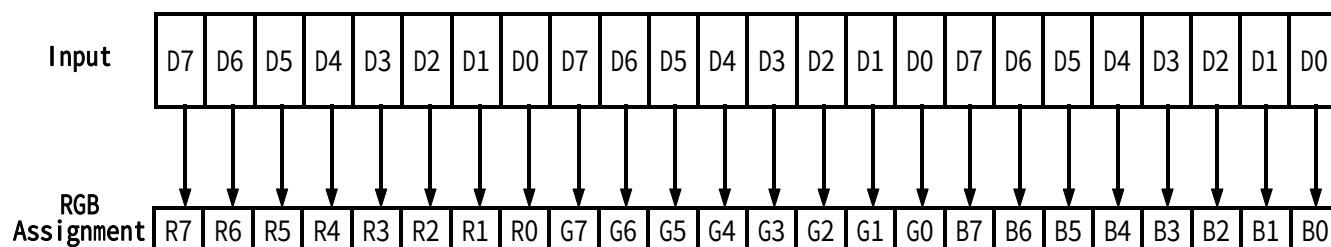
6. Operation Specifications

6.1 Serial Peripheral Interface



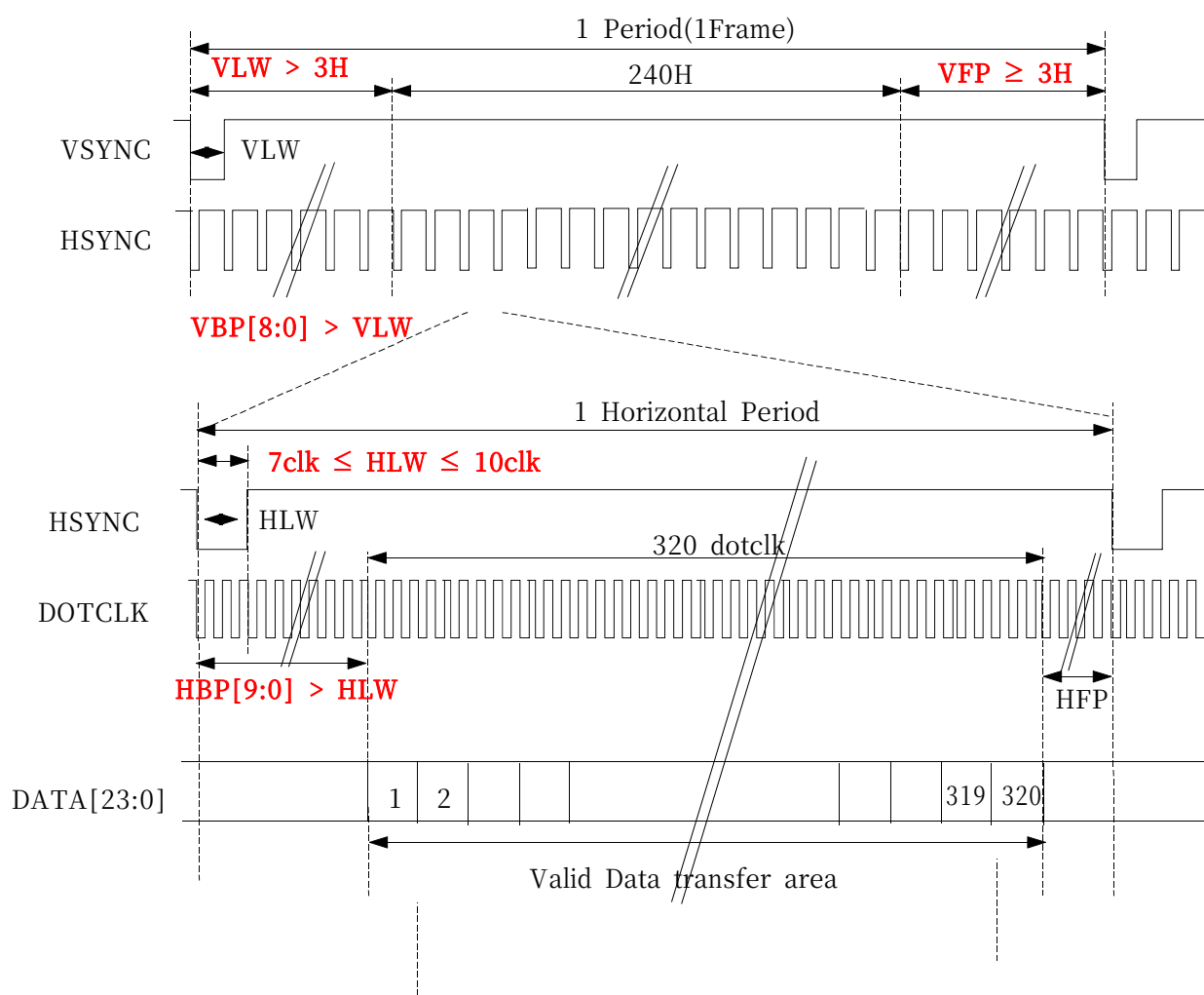
(Note) RS="0": Index data
RS="1": Instruction data

6.2 Data Format for 24bit RGB Interface



6.3 24bit RGB Interface Timing

a. SYNC mode



$$\begin{aligned} * \text{DOTCLK} &= f_{\text{frame}} \times (240 + VBP + VFP) \times (320 + HBP + HFP) \\ &= 60\text{Hz} \times (240 + VBP + VFP) \times (320 + HBP + HFP) \end{aligned}$$

$$* 3H \leq VFP$$

$$* HBP > HLW$$

$$* VBP > VLW$$

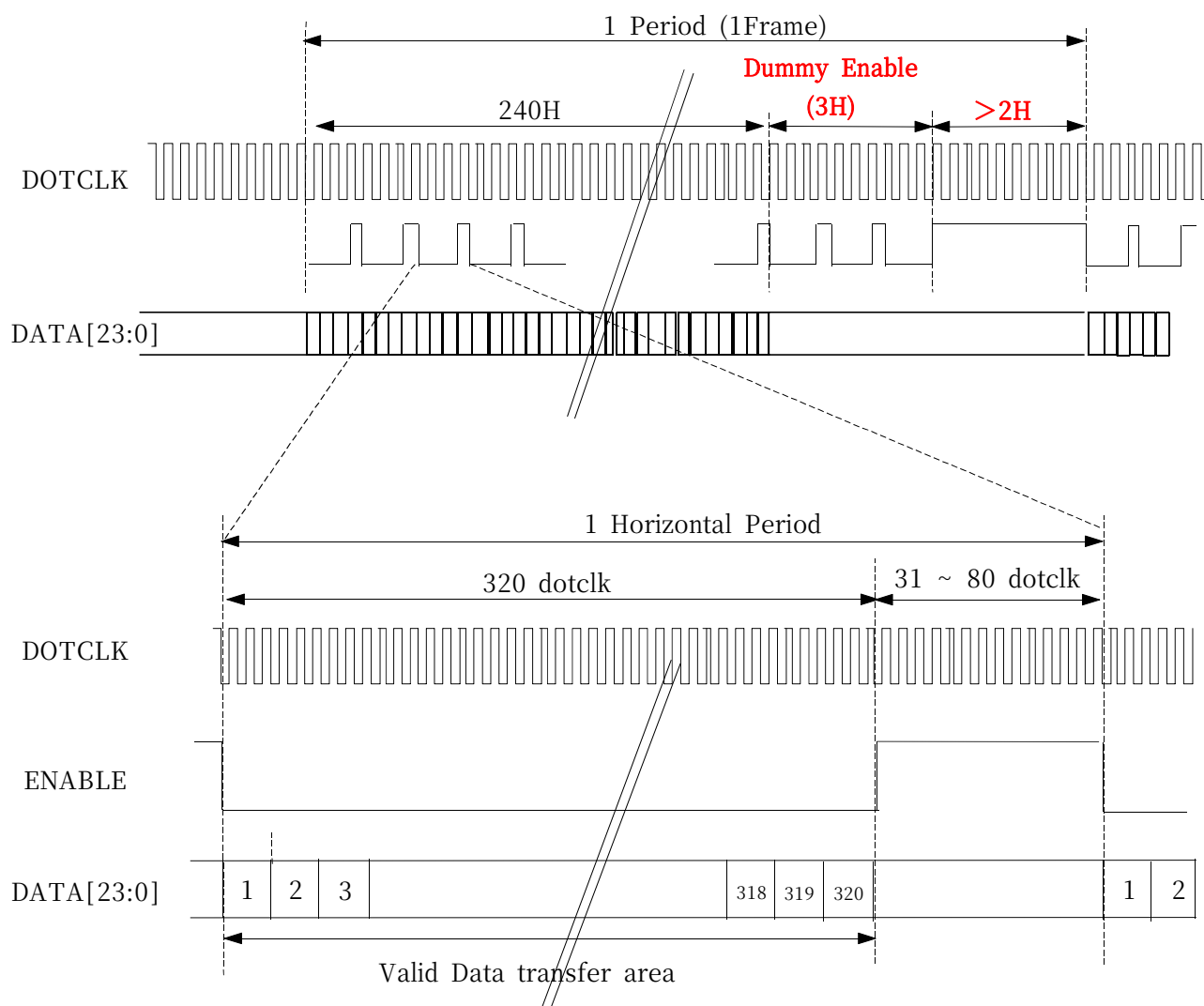
$$* 7clk \leq HLW \leq 10clk$$

$$* 3H \leq VLW$$

Note 1. Polarity register setting : VPL = 0, HPL = 0, DPL=0

2. The rising edge of DOTCLK is used to fetch display data

b. DE mode



$$\begin{aligned}
 * \text{ DOTCLK} &= f_{\text{frame}} \times (240 + \text{VBP} + \text{VFP}) \times (320 + \text{HBP} + \text{HFP}) \\
 &= 60\text{Hz} \times (240 + \text{VBP} + \text{VFP}) \times (320 + \text{HBP} + \text{HFP})
 \end{aligned}$$

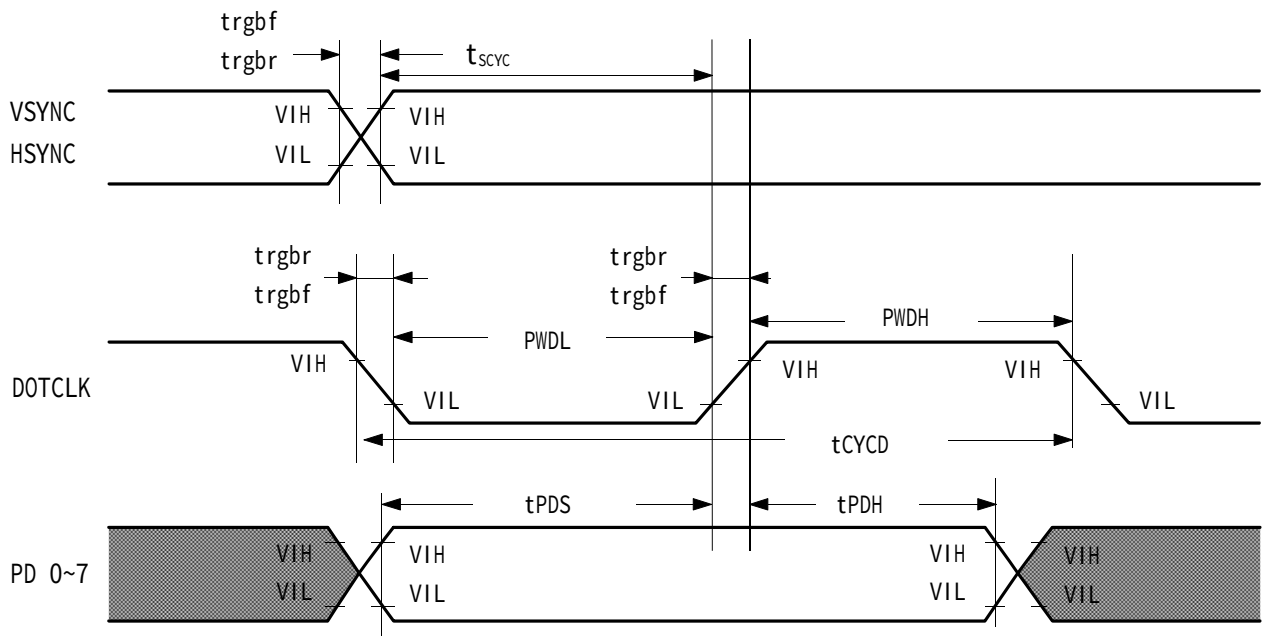
Note 1. Polarity register setting : EPL = 0, DPL=0

2. The rising edge of DOTCLK is used to fetch display data

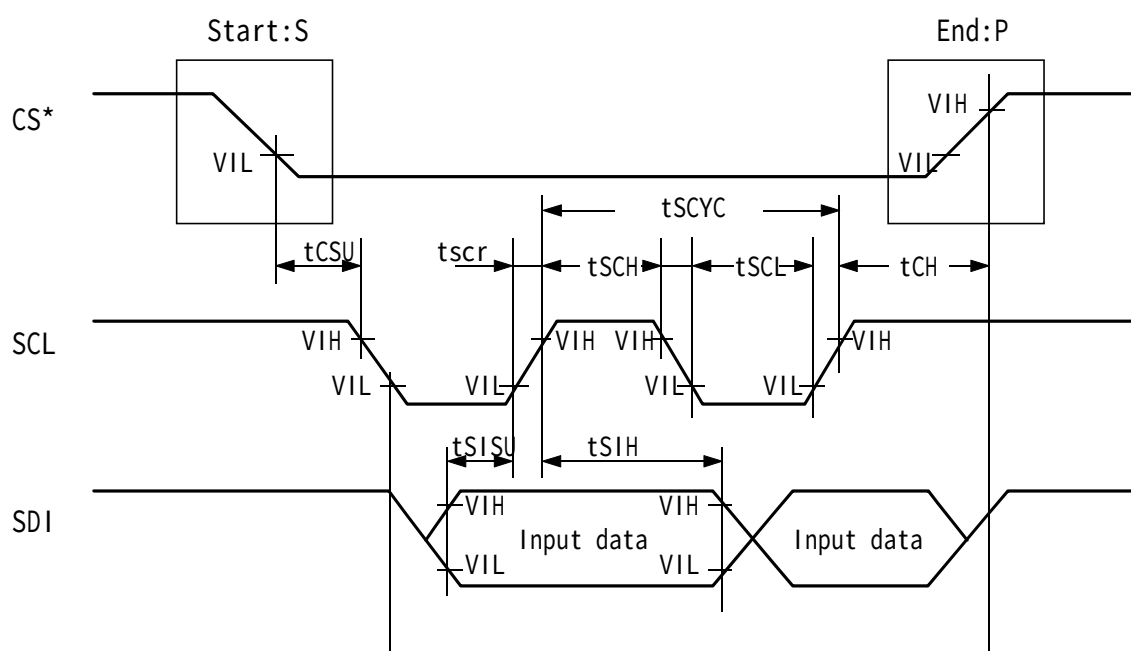
6.4 Electrical Specifications

Item	Symbol	Min	Typ	Max	Unit
VSYNC/HSYNC setup time	t_{SCYC}				ns
DOTCLK "L" level pulse width	P_{WDL}				ns
DOTCLK "H" level pulse width	P_{WDH}				ns
DOTCLK cycle time	t_{CYCLD}				ns
Data setup time	t_{PDS}				ns
Data hold time	t_{PDH}				ns
DOTCLK,VSYNC,HSYNC, rise/fall time	$trgbr, trgf$				ns

T.B.D.



Item		Symbol	Min	Typ	Max	us
Serial clock cycle time	Write (received)	t_{SCYC}				ns
Serial clock high-level pulse width	Write (received)	t_{SCH}	T.B.D.			ns
Serial clock low-level pulse width	Write (received)	t_{SCL}				ns
Serial clock rise/fall time		$t_{scr,scf}$				ns
Chip select setup time		t_{CSU}				ns
Chip select hold time		t_{CH}				ns
Serial input data setup time		t_{SISU}				ns



7. Command List & Instruction Descriptions

7.1 Command List

Refer to the Appendix 1: S6F2002 IC Specification

7.2 Instruction Descriptions

Refer to the Appendix 1: S6F2002 IC Specification

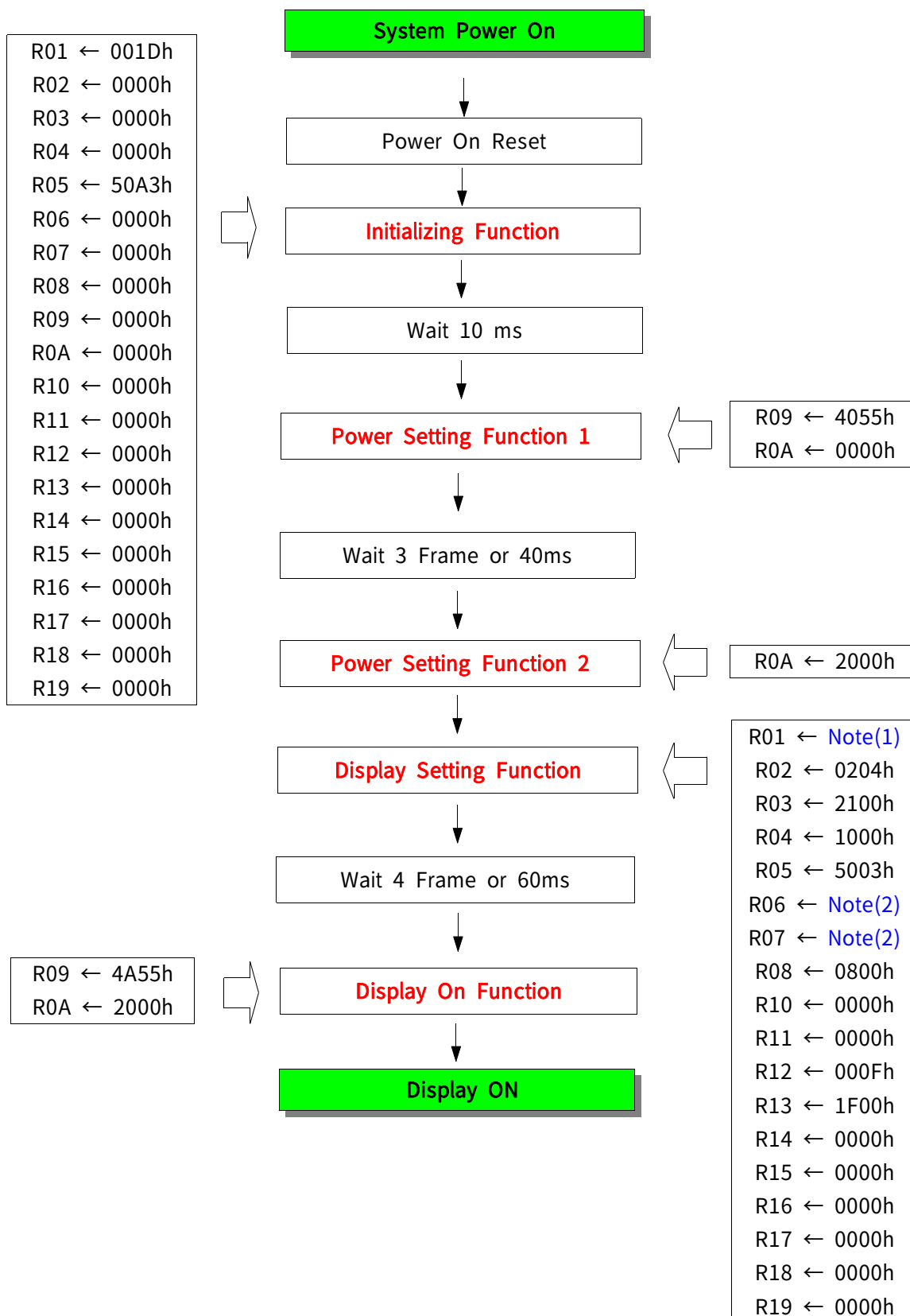
8. Reset

If the /RESET input becomes "L" or the reset command is input, the each register to its default value. These default values are listed in the table below.

Register		Default	Note
Display Interface Control	R01	001DH	-
Display Data Control	R02	0000H	-
Entry Mode	R03	0000H	-
Gate Control 1	R04	0000H	-
Gate Control 2	R05	0000H	-
Display Control 1	R06	50A3H	-
Display Control 1	R07	0000H	-
Source Output Timing Control	R08	0000H	-
Power Contorl 1	R09	0000H	-
Power Contorl 2	R0A	0000H	-
Gamma Control	R10 ~ R19	0000H	-

9. Power On/Off Sequence

9.1 Power On Sequence



< Note (1) >

- Register 'R01' determine specify the interface mode

(R01H)

IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
0	1	SSMD	0	0	0	0	0	0	0	0	1	1	1	0	1

- SSMD Bits Setting

SSMD	Interface	R01H
0	SYNC mode	409D
1	DE mode	609D

< Note (2) >

- Register 'R06', 'R07' determine Hsync Backporch('R06') & Vsync Backporch period('R07')

- Backporch limit

(1) 7clk < Hsync Backporch < 1024clk

(2) 3line < Vsync Backporch < 512line

(R06H)

IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
0	0	0	0	0	0	0	VBP8	VBP7	VBP6	VBP5	VBP4	VBP3	VBP2	VBP1	VBP0

(R07H)

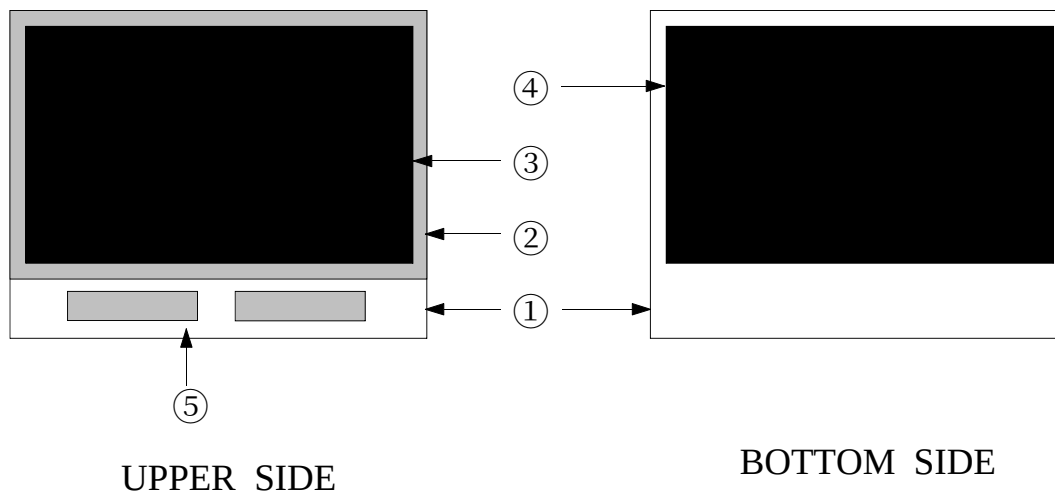
IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
0	0	0	0	0	0	0	HBP8	HBP7	HBP6	HBP5	HBP4	HBP3	HBP2	HBP1	HBP0

9.2 Power Off Sequence

T.B.D.

10. Part List of Components

10.1 Panel, Driver IC

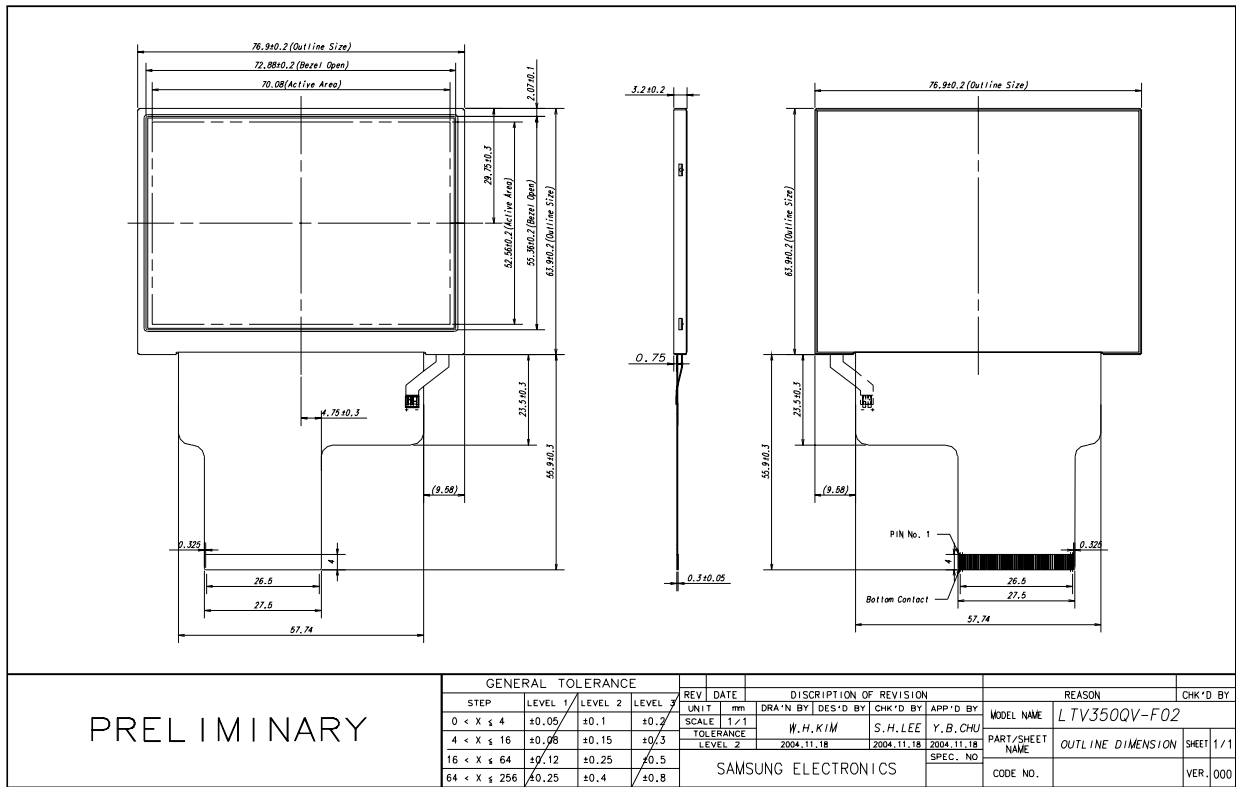


No.	Part Name	Specification	Maker	EA	Note
1	TFT GLASS	T0.5	SAMSUNG CORNING	1	①
2	COLOR FILTER GLASS	T0.5	SEC	1	②
3	UPPER POLARIZER	Thin-Pol	KORENO	1	③
4	LOWER POLARIZER	Thin-Pol	KORENO	1	④
5	DRIVER IC	S6F2002	LDI	2	⑤

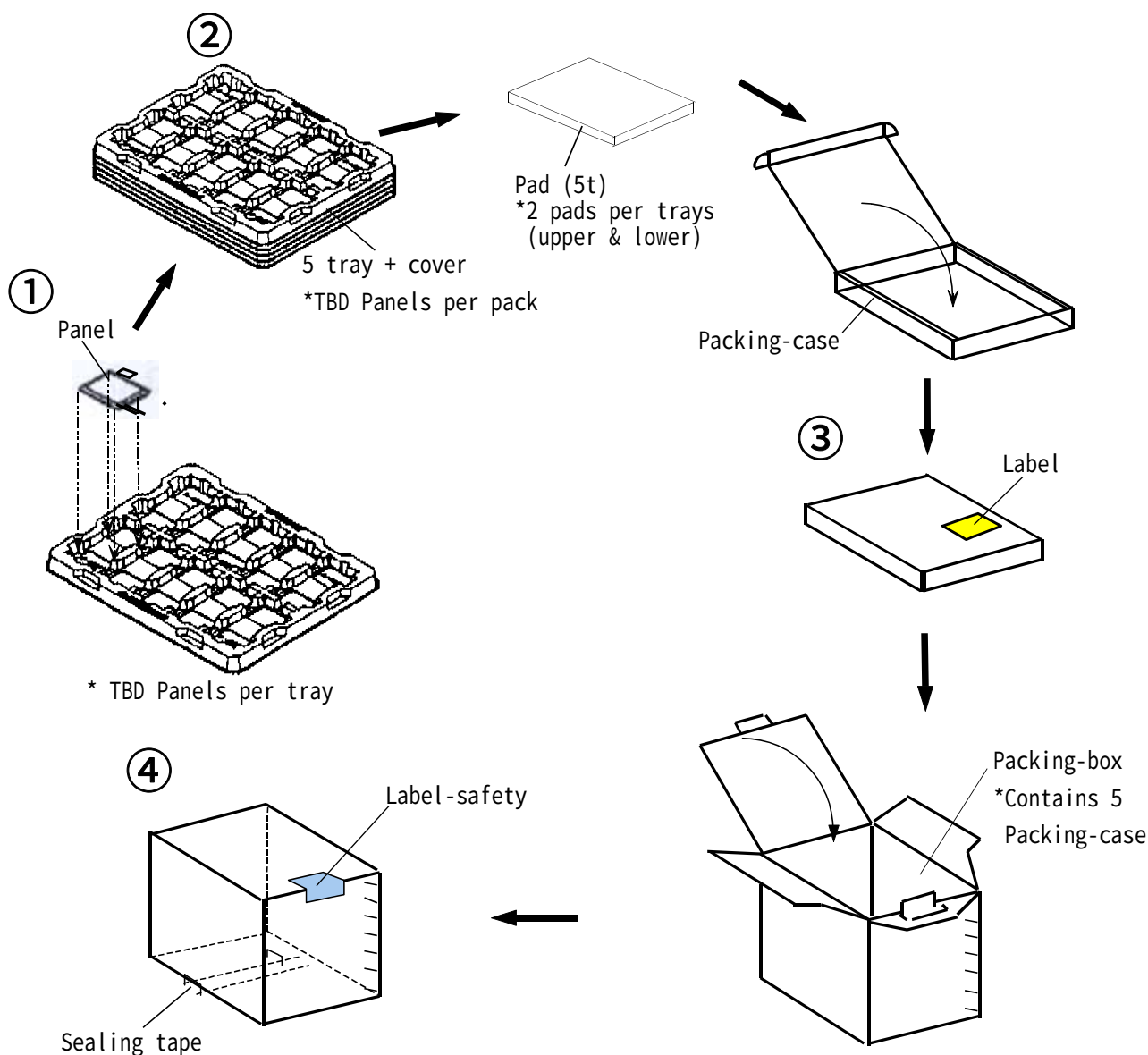
11. Reliability Test Result

T.B.D.

12. Outline Dimension



13. Packing



Note (1) Total : Case: Approx. **TBD** Kg

Box: Approx. **TBD** Kg

(2) Size : Case: 490(W) x 342(D) x 58(H)

Box: 505(W) x 355(D) x 312(H)

(3) Place the panels in the tray facing the direction shown in the figure.

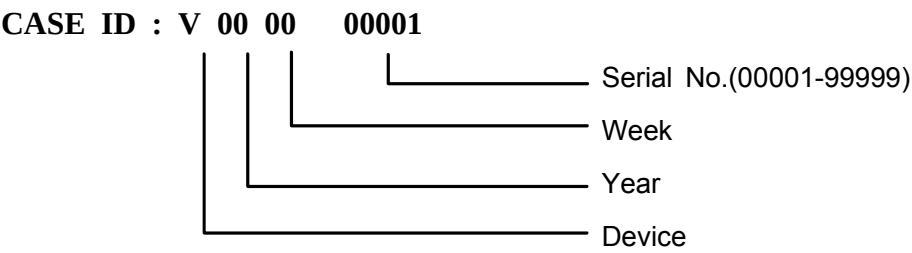
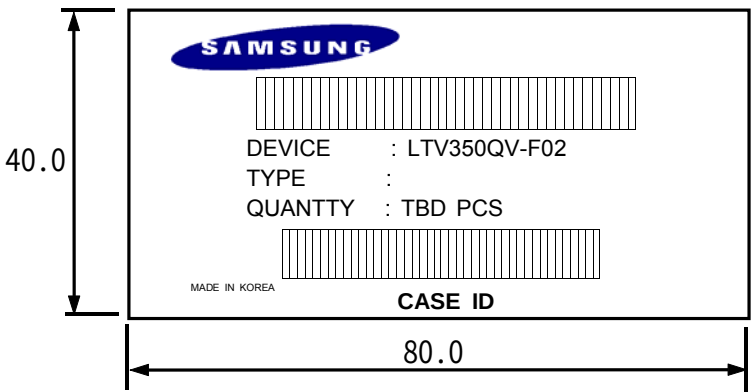
(4) Place 5 tray and cover(empty tray) and pads inside the packing-case.

(5) Place 5 packing-case inside the packing-box.(Affix the label)

(6) Seal the packing-box. Affix the label-safety.

14. Marking & Others

(1) Packing case attach



15. General Precautions

15.1 Handling

- (a) When the module is assembled, it should be attached to the system firmly. Be careful not to twist and bend the module.
- (b) Refrain from strong mechanical shock and / or any force to the module. In addition to damage, this may cause improper operation or damage to the module and back-light unit.
- (c) Note that polarizers are very fragile and could be easily damaged. Do not press or scratch the surface harder than a HB pencil lead.
- (d) Wipe off water droplets or oil immediately. If you leave the droplets for a long time, Staining and discoloration may occur.
- (e) If the surface of the polarizer is dirty, clean it using some absorbent cotton or soft cloth.
- (f) The desirable cleaners are water, IPA(Isopropyl Alcohol) or Hexane. Do not use Ketone type materials(ex. Acetone), Ethyl alcohol, Toluene, Ethyl acid or Methyl chloride. It might permanent damage to the polarizer due to chemical reaction.
- (g) If the liquid crystal material leaks from the panel, it should be kept away from the eyes or mouth. In case of contact with hands, legs or clothes, it must be washed away thoroughly with soap.
- (h) Protect the module from static , it may cause damage to the Integrated Gate Circuit.
- (i) Use finger-stalls with soft gloves in order to keep display clean during the incoming inspection and assembly process.
- (j) Do not disassemble the module.
- (k) Protection film for polarizer on the module shall be slowly peeled off just before use so that the electrostatic charge can be minimized.
- (l) Pins of I/F connector shall not be touched directly with bare hands

15.2 Storage

- (a) Do not leave the panel in high temperature, and high humidity for a long time. It is highly recommended to store the module with temperature from 0 to 35°C and relative humidity of less than 70%.
- (b) Do not store the TFT-LCD module in direct sunlight.
- (c) The module shall be stored in a dark place. It is prohibited to apply sunlight or fluorescent light during the store.

15.3 Operation

- (a) Do not connect, disconnect the module in the "Power On" condition.
- (b) Power supply should always be turned on/off by the "Power on/off sequence"

15.4 Others

- (a) The Liquid crystal is deteriorated by ultraviolet, do not leave it in direct sunlight and strong ultraviolet ray for many hours.
- (b) Avoid condensation of water. It may result in improper operation or disconnection of electrode.
- (c) Do not exceed the absolute maximum rating value. (the supply voltage variation, input voltage variation, variation in part contents and environmental temperature, and so on) Otherwise the panel may be damaged.
- (d) If the panel displays the same pattern continuously for a long period of time, it can be the situation when the image "Sticks" to the screen.
- (e) This panel has its circuitry FPC on the bottom side and should be handled carefully in order not to be stressed.